

An Inverse-Jet Atlas for q -Analogues

Multivariate Reversion, Singular Branches, and Asymptotics at
 $q = 0, 1, -1$, Roots of Unity, and $q = \pm\infty$

A second research companion to the ProveIt q -series drafts

August 30, 2026

Abstract

The ProvelIt draft corpus already contains a substantial forward theory of q -Pochhammer symbols, Gaussian coefficients, and their asymptotic expansions, together with a first report on inverse q -analogs. The present volume deliberately starts beyond that baseline. Its organizing object is an *inverse jet*: a local inverse expansion equipped with its recovered parameter, base point, ramification index, branch label, domain of validity, and dependence on every remaining parameter. Ordinary Taylor inverses, Puiseux inverses, logarithmic inverses, and multiscale inverses are treated in one framework.

Several results appear to be new within the audited repository corpus. For the finite product $P_n(a, q) = (a; q)_n$, the nontrivial parameter for which $q = -1$ becomes critical is exactly

$$a_* = \frac{1}{n-1} \quad (n \text{ even}), \quad a_* = -\frac{1}{n} \quad (n \text{ odd}).$$

For every $n \geq 4$ this produces a quadratic fold and hence two square-root inverse branches. The sole low-order exception is the exact cubic identity

$$(-1/3; q)_3 = \frac{32}{27} + \frac{4}{27}(q+1)^3,$$

which gives three cube-root branches without a remainder. At arbitrary roots of unity we express the ramification order by cyclotomic multiplicity. A Gaussian polynomial is square-free because each multiplicity is 0 or 1; a q -multinomial can have higher multiplicity, equal to an explicit residue-carry count. In particular,

$$[n]_q! = \lfloor n/2 \rfloor! (q+1)^{\lfloor n/2 \rfloor} (1 + \mathcal{O}(q+1))$$

near $q = -1$, so its inverse has a precisely normalized higher Puiseux branch.

For the infinite product we derive a fixed-target double-scaling inverse that is distinct from the usual fixed-argument $q \rightarrow 1$ expansion. If $(a; e^{-\varepsilon})_\infty = e^{-L}$, with $L > 0$ fixed, then

$$a = L\varepsilon - \frac{L(L+2)}{4}\varepsilon^2 + \frac{L(L^2+9L+12)}{72}\varepsilon^3 - \frac{L(L^3+4L^2+12L+24)}{576}\varepsilon^4 + \mathcal{O}(\varepsilon^5).$$

An arbitrary-order recurrence is supplied. For q -gamma we obtain an all-order algebraic expansion in which every odd power beyond the linear term vanishes. Reversion yields expansions both for the base and for the argument. For q -beta, the base inverse changes type on the explicit critical curve $xy = 1$: it is ordinary off the curve, square-root on the curve except at the identically constant point $(1, 1)$.

For Gaussian coefficients we prove a general centered-palindromic theorem. A nonconstant polynomial with nonnegative palindromic coefficients becomes, after centering, the moment-generating function of a symmetric finite distribution. Its normalized inverse therefore has exactly two global real branches, and its branch at $q = 1$ is governed by cumulants. This covers Gaussian and q -multinomial coefficients and many finite product analogs. The report also treats inverses of q -numbers, q -factorials, complex orders, q -gamma, q -digamma, q -beta, Euler q -exponentials, basic hypergeometric series, theta products, and the effective geometric base in Fabius–Rvachev formulas. Proved results, reproducible symbolic/numerical checks, and conjectures are explicitly separated.

Corpus audit and novelty boundary

The source snapshot is the main branch of Vladimir Reshetnikov’s `ProveIt` repository at commit `1cea73234a363ddbc392816f6babb5a57920e984`, dated 30 August 2026. The live subtree named in the research request contains the large synthesis *Exponents and q -Series Frontiers*, the Fabius parameter-response and Rvachev frontier reports, the forward expansion report, the proof-oriented q -Pochhammer/ q -binomial monograph, and the earlier inverse report, together with small generated TeX data tables. The code-search index lagged behind the live Contents tree, so the audit used the latter and the current commit tree rather than the stale search count.

The following material is treated as established repository prior art and is not repackaged as novelty here:

1. the finite/infinite product identities, Gaussian recurrences, q -gamma interpolation, and Bernoulli/poly-logarithmic forward expansions developed in the monograph and expansion report;
2. the first inverse report’s real branch classifications, principal inverse of $(a; q)_\infty$, positive-base uniqueness for Gaussian coefficients, secondary discriminants, and numerical branch tracking;
3. the Fabius–Rvachev use of the fixed half-base $q = 1/2$, geometric coefficient arrays, infinite sinc products, Bernoulli–Bell moment machinery, and parameter-response constructions.

The new emphasis is the missing *inverse atlas*: explicit inverse jets at all singular bases, systematic inversion in every parameter, and transport between regular and ramified regimes. “New” means not located in the active corpus under equivalent notation; it is not an unconditional claim of bibliographic priority.

Label	Meaning
PROVED	A proof is given here, or a precisely cited standard theorem is used with its hypotheses displayed.
SYMBOLIC/NUMERICAL	An exact computer-algebra identity or high-precision check is reproduced by the accompanying Python program.
CONJECTURAL	A precise conjecture or research direction; no proof is claimed.

How to read the atlas

An entry for an inverse branch records six pieces of data:

$$(F, p_0, y_0; m; \mathcal{B}; \mathcal{S}).$$

Here F is the forward function; p is the recovered parameter; (p_0, y_0) is the base point; m is the ramification order; $\mathcal{B}$ is the branch convention; and $\mathcal{S}$ is the singular set in all auxiliary parameters. Thus $m = 1$ means an ordinary Taylor inverse, $m > 1$ a Puiseux inverse, and $m = \infty$ is mnemonic for an essential or logarithmic inversion. An inverse without these data is normally ambiguous.

The variable

$$q = e^{-\varepsilon}$$

is used near $q = 1$. It preserves reciprocal symmetry: $q \mapsto q^{-1}$ becomes $\varepsilon \mapsto -\varepsilon$. Near infinity we use $Q = q^{-1}$. At a primitive d -th root of unity we write $q = \zeta + h$. The target is frequently logarithmically normalized:

$$w = \log \frac{y}{F(p_0)},$$

which converts products into additive cumulant jets.

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Part I

Universal inversion machinery

Chapter 1

Parameter-wise inverses and their geometry

1.1 One equation does not recover every parameter

Let $F(p_1, \dots, p_r) = y$ be a scalar q -analog. Unless additional relations are supplied, the level set is generically an $(r - 1)$ -dimensional hypersurface. Consequently an “inverse of F ” always means either a partial inverse in one named parameter or a simultaneous inverse of a square system.

Definition 1.1 (Partial inverse branch). Fix auxiliary parameters λ and write $F(p; \lambda)$. A partial inverse branch is a function $p = G(y; \lambda)$, together with a base point and continuation rule, such that

$$F(G(y; \lambda); \lambda) = y.$$

Its singular values are the images of points where $F_p = 0$, together with values induced by poles, essential singularities, or singular auxiliary parameters.

Theorem 1.1 (PROVED Local partial inverse and first sensitivities). *Suppose F is holomorphic near (p_0, λ_0) , put $y_0 = F(p_0, \lambda_0)$, and assume $F_p(p_0, \lambda_0) \neq 0$. Then there is a unique local holomorphic branch $p = G(y; \lambda)$ with $G(y_0; \lambda_0) = p_0$. On this branch,*

$$G_y = \frac{1}{F_p}, \quad G_{\lambda_j} = -\frac{F_{\lambda_j}}{F_p}.$$

Proof. The holomorphic implicit-function theorem gives existence and uniqueness. Differentiate $F(G(y; \lambda), \lambda) = y$ first in y and then in λ_j . $\square$

These identities are also condition-number formulas. A relative condition number for recovering p from y is

$$\kappa_{p \leftarrow y} = \left| \frac{y}{p F_p} \right|,$$

when $p y F_p \neq 0$. It diverges at a fold and therefore predicts the transition from Taylor to Puiseux inversion.

1.2 The inverse Riemann surface

For a fixed auxiliary parameter the analytic curve

$$\mathcal{R}_F = \{(p, y) \in \mathbb{C}^2 : F(p) = y\}$$

projects to the target plane. Away from critical values the projection is a covering. Loops in the punctured target plane permute the sheets; this monodromy is part of the inverse, not an implementation detail. For a finite q -Pochhammer product inverted in its argument, the covering has n sheets. Inverting in q gives degree $n(n-1)/2$ generically. Infinite products replace the algebraic covering by an infinite-sheeted analytic one.

1.3 Branch labels used below

Real branches are labelled by their monotonicity interval. Complex branches are labelled by a germ at a regular point and a homotopy class of continuation path. Near a ramification point of order m , branches are indexed by $\omega_m^j = \exp(2\pi i j/m)$, $0 \leq j < m$. Near infinity, a chosen branch of $y^{1/m}$ is always stated explicitly.

Chapter 2

Ordinary inverse jets to arbitrary order

2.1 Lagrange–Bürmann in coefficient form

Let

$$w = f(z) = c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad c_1 \neq 0.$$

Write $z = g(w) = \sum_{n \geq 1} b_n w^n$. Lagrange–Bürmann gives

$$b_n = \frac{1}{n} [z^{n-1}] \left(\frac{z}{f(z)} \right)^n. \quad (2.1)$$

The first coefficients are

$$b_1 = \frac{1}{c_1}, \quad (2.2)$$

$$b_2 = -\frac{c_2}{c_1^3}, \quad (2.3)$$

$$b_3 = \frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4}, \quad (2.4)$$

$$b_4 = -\frac{5c_2^3}{c_1^7} + \frac{5c_2 c_3}{c_1^6} - \frac{c_4}{c_1^5}. \quad (2.5)$$

Proposition 2.1 (PROVED Triangular reversion recurrence). *If $b_1 = 1/c_1$, then for $n \geq 2$,*

$$b_n = -\frac{1}{c_1} \sum_{k=2}^n c_k \sum_{\substack{j_1 + \cdots + j_k = n \\ j_i \geq 1}} b_{j_1} \cdots b_{j_k}.$$

This computes the inverse jet using only additions and multiplications in the coefficient ring with c_1^{-1} adjoined.

Proof. Extract the coefficient of w^n from $f(g(w)) = w$. The contribution of $c_1 g$ is $c_1 b_n$; every other term contains only b_j with $j < n$. $\square$

This recurrence is the default arbitrary-order mechanism throughout the report. It works equally well when each c_j is a rational function, polylogarithm, Bernoulli polynomial, or formal series in auxiliary parameters.

2.2 Bell-polynomial packaging

If the forward jet is exponentiated,

$$\frac{F(p_0 + z)}{F(p_0)} = \exp \left(\sum_{r \geq 1} \ell_r \frac{z^r}{r!} \right),$$

then its ordinary coefficients are complete exponential Bell polynomials:

$$[z^n] \frac{F(p_0 + z)}{F(p_0)} = \frac{1}{n!} B_n(\ell_1, \dots, \ell_n).$$

Reversion can therefore be organized as the composition

$$\text{logarithmic derivatives} \longrightarrow \text{Bell polynomial} \longrightarrow \text{triangular inverse recurrence}.$$

This is especially efficient for products because their logarithmic derivatives are short power sums.

2.3 Mixed parameter jets

For $F(p, \lambda) = y$, write $p = p_0 + \sum_{r+s \geq 1} b_{rs} u^r v^s$, where $u = y - y_0$ and $v = \lambda - \lambda_0$. Substitution into the bivariate Taylor series of F gives a triangular recurrence ordered by total degree. The first mixed derivatives are

$$G_{yy} = -\frac{F_{pp}}{F_p^3}, \tag{2.6}$$

$$G_{y\lambda} = -\frac{F_{p\lambda}}{F_p^2} + \frac{F_{pp}F_\lambda}{F_p^3}, \tag{2.7}$$

$$G_{\lambda\lambda} = -\frac{F_{\lambda\lambda}}{F_p} + 2\frac{F_{p\lambda}F_\lambda}{F_p^2} - \frac{F_{pp}F_\lambda^2}{F_p^3}. \tag{2.8}$$

They quantify how an inverse in one parameter deforms when another parameter moves, and they are indispensable near $q = 1/2$ in Fabius applications.

Chapter 3

Singular inverses: Puiseux, logarithmic, and multiscale regimes

3.1 Finite ramification

Suppose

$$F(p_0 + z) = y_0 + c_m z^m + c_{m+1} z^{m+1} + \cdots, \quad c_m \neq 0,$$

with $m \geq 2$. Set $\xi = ((y - y_0)/c_m)^{1/m}$. Newton–Puiseux gives m branches

$$p_j(y) = p_0 + \omega_m^j \xi - \frac{c_{m+1}}{m c_m} \omega_m^{2j} \xi^2 + \mathcal{O}(\xi^3), \quad 0 \leq j < m. \quad (3.1)$$

The expansion is convergent for algebraic functions and locally convergent for holomorphic F after choosing a slit neighborhood.

Proof of the displayed coefficient. Put $z = \omega_m^j \xi + A \xi^2 + \cdots$ and substitute into $(F - y_0)/c_m = \xi^m$. The coefficient of ξ^{m+1} is $m \omega_m^{j(m-1)} A + (c_{m+1}/c_m) \omega_m^{j(m+1)}$. Since $\omega_m^{jm} = 1$, solving gives the stated term. $\square$

3.2 Unfolding a fold

If $F_p = 0$, $F_{pp} \neq 0$, and an auxiliary parameter λ varies, then locally

$$y - y_* = A(p - p_*)^2 + B(\lambda - \lambda_*) + C(p - p_*)(\lambda - \lambda_*) + \cdots.$$

The discriminant curve in the (y, λ) -plane is obtained by eliminating p from $F - y = 0$ and $F_p = 0$. Crossing it exchanges zero, one, or two real branches. This is the local model for the $q = -1$ finite-product theorem proved later.

3.3 Essential and logarithmic inversion

At $q = 1$, infinite products often have the form

$$\log F(e^{-\varepsilon}) = -\frac{A}{\varepsilon} + B - C\varepsilon + D\varepsilon^3 + \cdots.$$

Let $X = -\log y + B$. Then

$$\varepsilon = \frac{A}{X} + \frac{A^2 C}{X^3} + \frac{A^4(-D) + 2A^3 C^2}{X^5} + \cdots, \quad (3.2)$$

with signs adjusted to the chosen forward convention. Exponentiating gives $q = e^{-\varepsilon}$. If ε also occurs inside a logarithm, Lambert W or transseries rather than a pure inverse-power series appears.

3.4 Newton polygons for multiple small scales

If the target and an auxiliary parameter approach a singular point together, a one-variable Taylor order is not intrinsic. Write

$$F(p_0 + z, \lambda_0 + v) - y_0 = \sum c_{rs} z^r v^s.$$

A face of the Newton polygon determines a balance $z \asymp v^\rho$. Each face polynomial supplies leading branches, after which a Puiseux recurrence continues them. The fixed-target inverse of $(a; q)_\infty$ near $q = 1$ is a particularly important double-scaling example: the correct balance is $a \asymp \varepsilon$, not fixed a .

Chapter 4

Simultaneous inversion and Good's multivariate formula

4.1 Square systems

Let $\mathbf{F} : \mathbb{C}^r \rightarrow \mathbb{C}^r$ and suppose $\det D\mathbf{F}(\mathbf{p}_0) \neq 0$. Then $\mathbf{p} = \mathbf{G}(\mathbf{y})$ exists locally and

$$D\mathbf{G} = (D\mathbf{F})^{-1}.$$

For example, recovering (a, q) from two measurements

$$y_1 = (a; q)_n, \quad y_2 = (aq^s; q)_m$$

is locally possible precisely when the two logarithmic sensitivity vectors are not parallel. This replaces the underdetermined scalar inverse by an identifiable two-channel experiment.

4.2 Good's multivariate Lagrange inversion

Suppose formal series w_i satisfy

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0, \quad 1 \leq i \leq r.$$

Good's formula [8] states that, for a formal series H ,

$$H(\mathbf{w}(t)) = \sum_{\mathbf{n} \geq \mathbf{0}} t^{\mathbf{n}} [z^{\mathbf{n}}] H(\mathbf{z}) \prod_{i=1}^r \phi_i(\mathbf{z})^{n_i} \det \left(\delta_{ij} - \frac{z_j}{\phi_i(\mathbf{z})} \frac{\partial \phi_i}{\partial z_j}(\mathbf{z}) \right)_{i,j=1}^r. \quad (4.1)$$

It is the natural coefficient engine when the recovered base, argument, and order are constrained by several q-product observations. The determinant is the multivariate Jacobian correction that has no scalar analogue.

4.3 A practical jet algebra

For computation, a multivariate jet is stored as a sparse map from multi-indices to exact coefficients. Composition is truncated by total or weighted degree; Newton iteration doubles the known order:

$$\mathbf{p}_{new} = \mathbf{p} - (D\mathbf{F}(\mathbf{p}))^{-1}(\mathbf{F}(\mathbf{p}) - \mathbf{y}).$$

Over rational-function coefficient fields this gives exact jets. Over floating-point balls it gives certified jets provided the Jacobian inverse and remainder bounds are enclosed.

Part II

Finite q -Pochhammer products

Chapter 5

Inversion in the argument

5.1 Forward algebra and logarithmic derivatives

For $n \in \mathbb{N}$, set

$$P_n(a, q) = (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad P_0 = 1.$$

The finite q -binomial theorem gives

$$P_n(a, q) = \sum_{m=0}^n (-1)^m q^{\binom{m}{2}} \begin{bmatrix} n \\ m \end{bmatrix}_q a^m. \quad (5.1)$$

Whenever no factor vanishes,

$$\partial_a^r \log P_n(a, q) = -(r-1)! \sum_{j=0}^{n-1} \frac{q^{jr}}{(1 - aq^j)^r}, \quad r \geq 1. \quad (5.2)$$

Thus every regular argument-inverse coefficient is an explicit rational function of finite geometric power sums.

5.2 The branch issuing from $a = 0$

Let $s = 1 - y$. From [Equation \(5.1\)](#),

$$s = [n]_q a - q \begin{bmatrix} n \\ 2 \end{bmatrix}_q a^2 + q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q a^3 + \cdots,$$

where $[n]_q = (1 - q^n)/(1 - q)$. Applying [Equation \(2.4\)](#) gives the following useful seed branch.

Theorem 5.1 ([PROVED](#) Argument inverse at the unit target). *If $[n]_q \neq 0$, the branch satisfying $a(1) = 0$ is*

$$\begin{aligned} a &= \frac{s}{[n]_q} + \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3} s^2 \\ &\quad + \left(\frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4} \right) s^3 + \mathcal{O}(s^4). \end{aligned} \quad (5.3)$$

To arbitrary order,

$$[s^m]a(s) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{1 - P_n(z, q)} \right)^m.$$

Proof. The displayed forward series has nonzero linear coefficient $[n]_q$. Apply Lagrange–Bürmann and then substitute the first three coefficients. $\square$

The singular set $[n]_q = 0$ is already informative: it consists of nontrivial n -th roots of unity. There the branch issuing from $a = 0$ ceases to be ordinary; the first nonzero elementary symmetric coefficient determines its Puiseux order.

5.3 Inverse jets at the exact zeros

For $q \neq 0$, the zeros in a are $a_j = q^{-j}$, $0 \leq j < n$. If they are distinct, each supplies an ordinary inverse branch at target $y = 0$.

Theorem 5.2 (PROVED Exact derivative at a geometric zero). *Assume $q^r \neq 1$ for $1 \leq r < n$. Then*

$$P_{n,a}(q^{-j}, q) = (-1)^{j+1} q^{-j(j-1)/2} (q; q)_j (q; q)_{n-1-j}. \quad (5.4)$$

Consequently the branch through $a_j = q^{-j}$ begins

$$a(y) = q^{-j} + \frac{(-1)^{j+1} q^{j(j-1)/2}}{(q; q)_j (q; q)_{n-1-j}} y + \mathcal{O}(y^2). \quad (5.5)$$

Proof. Factor the vanishing term:

$$P_{n,a}(q^{-j}, q) = -q^j \prod_{\ell \neq j} (1 - q^{\ell-j}).$$

For $\ell < j$, write $1 - q^{\ell-j} = -q^{-(j-\ell)}(1 - q^{j-\ell})$. Their product is $(-1)^j q^{-j(j+1)/2} (q; q)_j$; the remaining factors give $(q; q)_{n-1-j}$. Multiplication by $-q^j$ proves Equation (5.4), and the inverse-function theorem gives Equation (5.5). $\square$

The quadratic term follows without expanding the whole polynomial. If $R_j(a) = \prod_{\ell \neq j} (1 - aq^\ell)$, then

$$\frac{P_{n,aa}(a_j, q)}{P_{n,a}(a_j, q)} = 2 \frac{R'_j(a_j)}{R_j(a_j)} = -2 \sum_{\ell \neq j} \frac{q^\ell}{1 - q^{\ell-j}}.$$

Substitute this into $a = a_j + y/P_a - P_{aa}y^2/(2P_a^3) + \dots$.

5.4 Positive real base: complete real branch skeleton

For $0 < q < 1$, the zeros $1 < q^{-(n-1)} < \dots < q^{-1} < q^0 = 1$ are positive and simple. Rolle's theorem implies that every zero of $\partial_a P_n$ is real, simple, and lies strictly between consecutive zeros of P_n . Therefore the real inverse covering is obtained by cutting the a -axis at these critical points. The critical values determine how many real preimages a target possesses. This recovers the branch skeleton used in the earlier inverse report, but Equation (5.4) adds exact local normalizations at every zero.

5.5 Argument inverse at infinity

Let $N = \binom{n}{2}$. The two leading powers of [Equation \(5.1\)](#) are

$$P_n(a, q) = (-1)^n q^N a^n + (-1)^{n-1} q^{(n-1)(n-2)/2} [n]_q a^{n-1} + \mathcal{O}(a^{n-2}).$$

Hence, for a chosen n -th root $t = (y/((-1)^n q^N))^{1/n}$,

$$a = t + \frac{[n]_q}{nq^{n-1}} + \mathcal{O}(t^{-1}). \quad (5.6)$$

There are n branches, obtained by multiplying t by the n -th roots of unity. The complete descending series is obtained by substituting $a = t + \sum_{r \geq 0} c_r t^{-r}$ into [Equation \(5.1\)](#); its recurrence is triangular.

Chapter 6

Base inversion near $q = 0$ and $q = 1$

6.1 The small-base combinatorial jet

Separate the $j = 0$ factor:

$$P_n(a, q) = (1 - a) \prod_{j=1}^{n-1} (1 - aq^j).$$

The coefficient of q^r in the normalized product is the finite subset sum

$$[q^r] \frac{P_n(a, q)}{1 - a} = \sum_{\substack{S \subseteq \{1, \dots, n-1\} \\ \sum_{j \in S} j = r}} (-a)^{|S|}. \quad (6.1)$$

This supplies all forward and inverse coefficients without differentiation.

For $n \geq 2$ and $a \notin \{0, 1\}$, put

$$r = \frac{(1 - a) - y}{a(1 - a)}.$$

The leading relation is $r = q + \mathcal{O}(q^2)$, so the base inverse is ordinary.

Proposition 6.1 (PROVED Small-base inverse through cubic order). *For the branch $q \rightarrow 0$,*

$$n = 2 : \quad q = r, \quad \text{exactly,}$$

$$n = 3 : \quad q = r - r^2 + (2 + a)r^3 + \mathcal{O}(r^4),$$

$$n \geq 4 : \quad q = r - r^2 + (1 + a)r^3 + \mathcal{O}(r^4).$$

All later coefficients are obtained by applying the triangular reversion recurrence to Equation (6.1).

Proof. For $n = 3$, $r = q + q^2 - aq^3 + \mathcal{O}(q^4)$. For $n \geq 4$, the subsets contributing through degree three give $r = q + q^2 + (1 - a)q^3 + \mathcal{O}(q^4)$. Reversion proves the formulas. $\square$

The excluded cases are structural, not removable bookkeeping. At $a = 0$, $P_n \equiv 1$; at $a = 1$, $P_n \equiv 0$ because the $j = 0$ factor vanishes. For $n = 1$, the product is independent of q .

6.2 The logarithmic jet at $q = 1$

Set $q = e^{-\varepsilon}$, and let $S_m(n) = \sum_{j=0}^{n-1} j^m$. For $|a| < 1$, expansion of each logarithm and interchange of finite sums gives

Theorem 6.1 (PROVED Complete finite-product jet at $q = 1$).

$$\log P_n(a, e^{-\varepsilon}) = n \log(1 - a) + \sum_{m \geq 1} \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) S_m(n) \varepsilon^m. \quad (6.2)$$

The identity extends by analytic continuation in a on any simply connected domain avoiding the moving zero set.

Proof. Use

$$\log(1 - ae^{-j\varepsilon}) = - \sum_{r \geq 1} \frac{a^r}{r} e^{-jr\varepsilon}$$

and expand the exponential. The coefficient of ε^m is $-1)^{m+1} \sum_r a^r r^{m-1} S_m(n)/m!$, which is the displayed polylogarithm. $\square$

Let

$$w = \log \frac{y}{(1-a)^n} = c_1 \varepsilon + c_2 \varepsilon^2 + c_3 \varepsilon^3 + \cdots, \quad c_m = \frac{(-1)^{m+1}}{m!} \text{Li}_{1-m}(a) S_m(n).$$

If $c_1 = S_1(n)a/(1-a) \neq 0$, then

$$\varepsilon = \frac{w}{c_1} - \frac{c_2}{c_1^3} w^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) w^3 + \mathcal{O}(w^4), \quad q = e^{-\varepsilon}. \quad (6.3)$$

The formula is uniform on compact auxiliary-parameter sets on which c_1 stays away from zero.

6.3 Inverting the argument while $q \rightarrow 1$

A different inverse problem keeps the target y fixed and recovers a . Let

$$a_0 = 1 - y^{1/n},$$

with a chosen branch of the n -th root. Solving $P_n(a(\varepsilon), e^{-\varepsilon}) = y$ using Equation (6.2) gives the following mixed jet.

Theorem 6.2 (PROVED Fixed-target argument inverse near $q = 1$). If $a_0 \neq 1$, then

$$\begin{aligned} a(\varepsilon) &= a_0 + \frac{n-1}{2} a_0 \varepsilon \\ &\quad + \frac{a_0(n-1)(2n-4-3a_0(n-1))}{24(1-a_0)} \varepsilon^2 + \mathcal{O}(\varepsilon^3). \end{aligned} \quad (6.4)$$

The higher coefficients are rational functions of a_0, n generated triangularly from Equation (6.2).

Proof. Substitute $a = a_0 + A\varepsilon + B\varepsilon^2 + \cdots$ into the logarithm of the defining equation. The constant term is $n \log(1 - a_0) = \log y$. The linear equation gives $A = (n-1)a_0/2$; the quadratic equation gives the stated B . $\square$

Chapter 7

The inverse atlas at $q = -1$ and at roots of unity

7.1 Generic regularity at $q = -1$

At the alternating base,

$$P_n(a, -1) = (1 - a)^{\lceil n/2 \rceil} (1 + a)^{\lfloor n/2 \rfloor}. \quad (7.1)$$

Assume temporarily $a \notin \{0, \pm 1\}$. Put

$$E = \sum_{\substack{1 \leq j < n \\ j \text{ even}}} j, \quad O = \sum_{\substack{1 \leq j < n \\ j \text{ odd}}} j.$$

Then

$$\partial_q \log P_n(a, q)|_{q=-1} = a \left(\frac{E}{1-a} - \frac{O}{1+a} \right). \quad (7.2)$$

Here $E = e(e+1)$, $e = \lfloor (n-1)/2 \rfloor$, and $O = o^2$, $o = \lceil (n-1)/2 \rceil$. Unless the right side vanishes, the inverse $q(y)$ is an ordinary Taylor series based at $y_0 = P_n(a, -1)$.

7.2 The unique nontrivial critical parameter

Theorem 7.1 (**PROVED** Complete finite-product criticality at $q = -1$). *Let $n \geq 2$ and $a \notin \{0, \pm 1\}$. Then $\partial_q P_n(a, -1) = 0$ if and only if*

$$a = a_* = \begin{cases} \frac{1}{n-1}, & n \text{ even}, \\ -\frac{1}{n}, & n \text{ odd}. \end{cases} \quad (7.3)$$

Write $n = 2s$ or $n = 2s + 1$. At this parameter,

$$\partial_q^2 \log P_{2s}(a_*, q)|_{q=-1} = -\frac{(s+1)(2s-1)^2}{12s(s-1)}, \quad s \geq 2, \quad (7.4)$$

$$\partial_q^2 \log P_{2s+1}(a_*, q)|_{q=-1} = -\frac{(s-1)(2s+1)^2}{12s(s+1)}, \quad s \geq 1. \quad (7.5)$$

Thus every case $n \geq 4$ is a strict quadratic fold. The case $n = 3$ is cubic and satisfies the exact identity

$$P_3(-1/3, q) = \frac{32}{27} + \frac{4}{27}(q+1)^3. \quad (7.6)$$

Proof. For $a \neq 0$, equation Equation (7.2) vanishes precisely when $E(1 + a) = O(1 - a)$, hence $a = (O - E)/(O + E)$. If $n = 2s$, then $E = s(s - 1)$, $O = s^2$, giving $a = 1/(2s - 1)$. If $n = 2s + 1$, then $E = s(s + 1)$, $O = s^2$, giving $a = -1/(2s + 1)$.

Differentiate

$$\partial_q \log P_n = -a \sum_{j=1}^{n-1} \frac{j q^{j-1}}{1 - a q^j}$$

once more. Separate even and odd j and use the polynomial sums of $j(j - 1)$ and j^2 . Substitution of Equation (7.3) simplifies to Equation (7.4) and Equation (7.5). The even expression is nonzero for $s \geq 2$; the odd expression is nonzero for $s \geq 2$. When $s = 1$, direct multiplication gives Equation (7.6). $\square$

Corollary 7.1 (PROVED Explicit local inverse types). *Let $y_* = P_n(a_*, -1)$. For $n \geq 4$, put L_2 equal to the appropriate quantity in Equations (7.4) and (7.5). The two branches are*

$$q = -1 \pm \sqrt{\frac{2(y - y_*)}{y_* L_2}} + \mathcal{O}(y - y_*). \quad (7.7)$$

Since $y_* > 0$ and $L_2 < 0$, the two nearby real branches occur for $y < y_*$. For $n = 3$,

$$q = -1 + \omega_3^j \left(\frac{27}{4} \left(y - \frac{32}{27} \right) \right)^{1/3}, \quad j = 0, 1, 2, \quad (7.8)$$

exactly.

7.3 The singular endpoint parameters $a = \pm 1$

The parameter $a = 1$ makes $P_n \equiv 0$ because the $j = 0$ factor vanishes; no scalar base inverse is identifiable. For $a = -1$, however, only the odd factors vanish at $q = -1$. Let $m = \lfloor n/2 \rfloor$. Then

Theorem 7.2 (PROVED Higher ramification of $(-1; q)_n$).

$$(-1; q)_n = 2^{\lfloor n/2 \rfloor} (2m - 1)!! (q + 1)^m (1 + \mathcal{O}(q + 1)). \quad (7.9)$$

Consequently the inverse at target zero has m Puiseux branches,

$$q = -1 + \omega_m^j \left(\frac{y}{2^{\lfloor n/2 \rfloor} (2m - 1)!!} \right)^{1/m} + \mathcal{O}(y^{2/m}).$$

Proof. For odd $j = 2r - 1$, $1 + q^j = (2r - 1)(q + 1) + \mathcal{O}((q + 1)^2)$. Every even factor, including $j = 0$, tends to 2. Multiply the leading factors. $\square$

7.4 Arbitrary roots of unity

Let ζ be a root of unity and assume $a \neq 1$. A factor with $j > 0$ vanishes at $q = \zeta$ exactly when $a\zeta^j = 1$. Each such factor has a simple zero in q . Hence

$$\text{ord}_{q=\zeta} P_n(a, q) = \#\{j \in \{1, \dots, n - 1\} : a\zeta^j = 1\}. \quad (7.10)$$

If the count is $m > 0$, the inverse at target zero has m branches of the form Equation (3.1). If the count is zero but $\partial_q P_n(a, \zeta) = 0$, the target is a nonzero critical value; its ramification is found from the first nonzero derivative. Eliminating a from $P_q = 0$, $P_{qq} = 0$, and higher equations produces the root-of-unity critical strata.

Chapter 8

Base inversion at $q = \pm\infty$

8.1 Reciprocal factorization

For $n \geq 2$, $a \notin \{0, 1\}$, and $N = \binom{n}{2}$, factor every term with $j \geq 1$:

$$P_n(a, q) = C(a)q^N \prod_{j=1}^{n-1} (1 - a^{-1}q^{-j}), \quad C(a) = (1 - a)(-a)^{n-1}. \quad (8.1)$$

Thus infinity is an ordinary point after the ramified coordinate $Y = (y/C)^{1/N}$ is introduced.

Theorem 8.1 (PROVED First large-base inverse terms). *On any chosen N -th-root branch,*

$$q = Y + \frac{1}{aN} + \mathcal{O}(Y^{-1}), \quad Y = \left(\frac{y}{C(a)} \right)^{1/N}. \quad (8.2)$$

The full expansion is a convergent Laurent series in Y^{-1} sufficiently near infinity on the corresponding sheet.

Proof. The product in Equation (8.1) equals $1 - a^{-1}q^{-1} + \mathcal{O}(q^{-2})$. Put $q = Y + c + \mathcal{O}(Y^{-1})$. Comparing the coefficient of Y^{-1} in $(q/Y)^N(1 - a^{-1}q^{-1} + \dots) = 1$ gives $Nc - a^{-1} = 0$. $\square$

For $q \rightarrow -\infty$, the same formula applies with a branch of $Y = (y/C)^{1/N}$ tending to the negative real axis. Whether a real branch is available depends on the sign of $C(-1)^N$ and on the target direction. The complex atlas is uniform: all N branches are related by roots of unity.

8.2 Reciprocal transport as an algorithm

Rather than expanding a high-degree polynomial at infinity directly, compute the small- Q product

$$R(Q) = \prod_{j=1}^{n-1} (1 - a^{-1}Q^j),$$

revert $Y = qR(1/q)^{1/N}$, and finally set $q = Q^{-1}$. This reuses the small-base subset-sum machinery and prevents catastrophic cancellation.

Chapter 9

Order inversion, multiple arguments, and mixed recovery

9.1 Complex order

For $0 < |q| < 1$, define

$$(a; q)_\alpha = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \quad (9.1)$$

with a chosen branch of $q^\alpha = e^{\alpha \text{Log } q}$. Its Lambert-series form is

$$\log(a; q)_\alpha = - \sum_{r \geq 1} \frac{a^r (1 - q^{\alpha r})}{r(1 - q^r)}. \quad (9.2)$$

Theorem 9.1 (PROVED All order derivatives). *On every domain of absolute convergence,*

$$\partial_\alpha^m \log(a; q)_\alpha = (\text{Log } q)^m \sum_{r \geq 1} \frac{r^{m-1} a^r q^{\alpha r}}{1 - q^r}, \quad m \geq 1. \quad (9.3)$$

Proof. Differentiate Equation (9.2) term by term. □

If the first derivative is nonzero, every coefficient of the local inverse $\alpha(y)$ follows from Equation (2.1) and Equation (9.3). If it vanishes, the first nonzero higher Lambert sum determines a Puiseux branch. The branch also depends on the logarithm used in q^α ; shifting $\text{Log } q$ by $2\pi i$ changes the global order covering.

9.2 Near $q = 1$: Bernoulli-polynomial order jet

The generating identity

$$\frac{1 - e^{-\alpha x}}{1 - e^{-x}} = \sum_{j \geq 0} \frac{B_{j+1}(1) - B_{j+1}(1 - \alpha)}{(j+1)!} x^j \quad (9.4)$$

gives, for fixed $|a| < 1$,

$$\log(a; e^{-\varepsilon})_\alpha = - \sum_{j \geq 0} \frac{B_{j+1}(1) - B_{j+1}(1 - \alpha)}{(j+1)!} \text{Li}_{1-j}(a) \varepsilon^j. \quad (9.5)$$

The leading term is $\alpha \log(1 - a)$, as required. If the target y is fixed and

$$\alpha_0 = \frac{\text{Log } y}{\text{Log}(1 - a)},$$

then, provided the chosen logarithms are consistent,

$$\alpha(\varepsilon) = \alpha_0 - \frac{\alpha_0(\alpha_0 - 1) \operatorname{Li}_0(a)}{2 \operatorname{Log}(1 - a)} \varepsilon + \mathcal{O}(\varepsilon^2). \quad (9.6)$$

The coefficient is valid for complex α_0 ; no integrality is used.

9.3 Discrete order and bracketing

For integer order,

$$\frac{(a; q)_{n+1}}{(a; q)_n} = 1 - aq^n.$$

On a real sign domain where these ratios stay in $(0, 1)$ or in $(1, \infty)$, the sequence is monotone and a discrete inverse order is located by binary search. The continuous inverse [Equation \(9.1\)](#) then supplies a sub-integer interpolation and a Newton correction. Outside such domains, a discrete inverse is naturally set-valued.

9.4 Several arguments and identifiability

For

$$F(a_1, \dots, a_s; q) = \prod_{i=1}^s (a_i; q)_{n_i},$$

one scalar target leaves an $(s - 1)$ -dimensional level set. If all but a_r are fixed, divide by the known factors and reuse the one-argument inverse. If several arguments are to be recovered, use at least as many shifted-product observations as unknowns. The Jacobian entries are the finite sums in [Equation \(5.2\)](#); their determinant is an exact identifiability criterion.

Part III

The infinite product and complex-order limits

Chapter 10

Argument inversion for $(a; q)_\infty$

10.1 The principal real branch

Assume $0 < q < 1$ and define

$$P_\infty(a, q) = (a; q)_\infty = \prod_{j \geq 0} (1 - aq^j).$$

For $a < 1$, all factors are positive and

$$\partial_a \log P_\infty(a, q) = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j} < 0.$$

Moreover $P_\infty(a, q) \rightarrow 0$ as $a \uparrow 1$, while it tends to infinity as $a \rightarrow -\infty$. Hence there is a unique real inverse

$$\mathcal{A}_q : (0, \infty) \longrightarrow (-\infty, 1), \quad (\mathcal{A}_q(y); q)_\infty = y.$$

This is the principal inverse used in the earlier report. The local formulas below refine its endpoint normalization and extend directly to nonprincipal complex branches.

10.2 The branch at $a = 0$

Euler's expansion is

$$(a; q)_\infty = \sum_{m \geq 0} \frac{(-1)^m q^{\binom{m}{2}}}{(q; q)_m} a^m. \quad (10.1)$$

Let $s = 1 - y$. Reversion gives

$$\begin{aligned} a &= (1 - q)s + \frac{q(1 - q)}{1 + q} s^2 \\ &\quad + (1 - q) \left(\frac{2q^2}{(1 + q)^2} - \frac{q^3}{(1 + q)(1 + q + q^2)} \right) s^3 + \mathcal{O}(s^4). \end{aligned} \quad (10.2)$$

The arbitrary coefficient is again

$$[s^m]a(s) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{1 - (z; q)_\infty} \right)^m.$$

The radius is limited by the nearest critical value of the entire function, not merely by the nearest zero.

10.3 All zero branches

The zeros are $a_m = q^{-m}$, $m \geq 0$, and are simple when q is not a root of unity. The finite-product calculation passes to the limit.

Theorem 10.1 (PROVED Exact derivative at every infinite-product zero). *For $0 < |q| < 1$ and $m \geq 0$,*

$$\partial_a(a; q)_\infty|_{a=q^{-m}} = (-1)^{m+1} q^{-m(m-1)/2} (q; q)_m (q; q)_\infty. \quad (10.3)$$

Thus

$$a(y) = q^{-m} + \frac{(-1)^{m+1} q^{m(m-1)/2}}{(q; q)_m (q; q)_\infty} y + \mathcal{O}(y^2)$$

on the branch through q^{-m} .

Proof. Factor the zero at index m . The indices below m produce $(-1)^m q^{-m(m+1)/2} (q; q)_m$, the indices above it produce $(q; q)_\infty$, and differentiation of $1 - aq^m$ contributes $-q^m$. $\square$

10.4 The principal branch at target zero

Put $a = 1 - \delta$. Since

$$(a; q)_\infty = (1 - a)(aq; q)_\infty,$$

and $(q; q)_\infty \neq 0$, the principal inverse has an ordinary endpoint jet. Let

$$L_1(q) = \sum_{j \geq 1} \frac{q^j}{1 - q^j}.$$

Then

$$\mathcal{A}_q(y) = 1 - \frac{y}{(q; q)_\infty} + L_1(q) \left(\frac{y}{(q; q)_\infty} \right)^2 + \mathcal{O}(y^3). \quad (10.4)$$

Higher coefficients are polynomials in the Lambert sums $\sum_{j \geq 1} q^{rj} / (1 - q^j)^r$.

10.5 Nonprincipal critical points

Critical points solve

$$\sum_{j \geq 0} \frac{q^j}{1 - aq^j} = 0.$$

They lie between successive positive zeros on the real axis for $0 < q < 1$, and each simple critical point creates a square-root pair in the target plane. The earlier inverse report develops their large-index asymptotics and log-periodic organization. The atlas viewpoint adds a practical rule: store each critical germ together with the exact zero interval (q^{-m}, q^{-m-1}) and the continuation path, because ordering by target value can change at a secondary-discriminant collision.

Chapter 11

Base inversion for the infinite product

11.1 At $q = 0$

For fixed $a \notin \{0, 1\}$, the first coefficients of the infinite product coincide with those of every finite product of order at least four. With

$$r = \frac{(1-a) - y}{a(1-a)},$$

we therefore have

$$q = r - r^2 + (1+a)r^3 + \mathcal{O}(r^4). \quad (11.1)$$

All orders follow from the subset-sum expansion over positive integers:

$$[q^N] \frac{(a; q)_\infty}{1-a} = \sum_{\substack{S \subset \mathbb{N}_{>0} \text{ finite} \\ \text{sum } S = N}} (-a)^{|S|}.$$

This is a weighted distinct-partition polynomial in a .

11.2 Fixed argument and the essential singularity at $q = 1$

For fixed $|a| < 1$, set $q = e^{-\varepsilon}$. The Lambert series

$$\log(a; q)_\infty = - \sum_{r \geq 1} \frac{a^r}{r(1-q^r)}$$

and the Bernoulli expansion of $(1 - e^{-x})^{-1}$ give

Theorem 11.1 (PROVED McIntosh-type logarithmic expansion).

$$\log(a; e^{-\varepsilon})_\infty = - \frac{\text{Li}_2(a)}{\varepsilon} + \frac{1}{2} \log(1-a) - \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} \text{Li}_{2-2m}(a) \varepsilon^{2m-1}. \quad (11.2)$$

The equality is interpreted as an asymptotic expansion in sectors avoiding the Stokes directions and the singularities of the chosen polylogarithm branches.

Proof. Insert

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} x^{2m-1}$$

into the Lambert series and sum powers of r as polylogarithms. Standard remainder estimates justify termwise truncation on compact subsets of $|a| < 1$; analytic continuation gives the sectorial form. This is the classical asymptotic mechanism developed by McIntosh [9]. $\square$

For $a \in (0, 1)$, write

$$A = \text{Li}_2(a), \quad B = \frac{1}{2} \log(1 - a), \quad C = \frac{1}{12} \text{Li}_0(a), \quad E = \frac{1}{720} \text{Li}_{-2}(a),$$

and $X = -\log y + B$. Then $y \rightarrow 0^+$ corresponds to $X \rightarrow \infty$, and reversion of [Equation \(11.2\)](#) gives

$$\varepsilon = \frac{A}{X} + \frac{A^2 C}{X^3} + \frac{2A^3 C^2 - A^4 E}{X^5} + \mathcal{O}(X^{-7}), \quad q = e^{-\varepsilon}. \quad (11.3)$$

This is the base inverse on the branch approaching $q = 1^-$. Other complex branches are obtained by the appropriate logarithmic and polylogarithmic continuations.

11.3 Monotonicity on principal sign domains

If $0 < a < 1$ and $0 < q < 1$, then

$$\partial_q \log(a; q)_\infty = -a \sum_{j \geq 1} \frac{j q^{j-1}}{1 - a q^j} < 0.$$

Thus the product decreases from $1 - a$ at $q = 0$ to 0 at $q = 1^-$, and for each $0 < y < 1 - a$ there is one positive real base. If $a < 0$, the same derivative is positive, so the product increases from $1 - a$ to infinity and again has one positive real base in its range. These global facts select the principal branch to which [Equation \(11.1\)](#) and [Equation \(11.3\)](#) belong.

Chapter 12

The fixed-target double-scaling inverse at

$$q = 1$$

12.1 Why fixed argument is the wrong scale

For fixed $a \in (0, 1)$, Equation (11.2) forces $(a; q)_\infty \rightarrow 0$ exponentially as $q \rightarrow 1^-$. To keep a target $y \in (0, 1)$ fixed, the argument must instead approach zero. Balancing the leading terms gives $a \asymp \varepsilon$. This regime is not obtained by simply substituting into a fixed- a inverse after truncation; the infinitely many powers of a and ε interact.

12.2 Explicit expansion

Let $L = -\log y$, with $L > 0$, and seek

$$a(\varepsilon) = \sum_{m \geq 1} A_m(L) \varepsilon^m.$$

Theorem 12.1 (PROVED Five-term fixed-target inverse). *The principal solution of*

$$(a; e^{-\varepsilon})_\infty = e^{-L}$$

has the expansion

$$\begin{aligned} a = L\varepsilon - \frac{L(L+2)}{4}\varepsilon^2 + \frac{L(L^2+9L+12)}{72}\varepsilon^3 \\ - \frac{L(L^3+4L^2+12L+24)}{576}\varepsilon^4 \\ - \frac{L(31L^4-75L^3-800L^2-720)}{86400}\varepsilon^5 + \mathcal{O}(\varepsilon^6). \end{aligned} \quad (12.1)$$

Every $A_m(L)$ is a polynomial in L with rational coefficients and a factor L .

Proof. Start from the exact Lambert series

$$\log(a; e^{-\varepsilon})_\infty = - \sum_{r \geq 1} \frac{a^r}{r(1 - e^{-r\varepsilon})}.$$

Insert the Bernoulli expansion of the denominator and the ansatz for a . At coefficient ε^{m-1} , the new unknown A_m occurs only in the term $-a/\varepsilon$, with coefficient -1 . The coefficient equation is therefore triangular. Solving successively gives the displayed polynomials. The same argument proves rational polynomiality and the factor L . $\square$

Algorithm 12.1 (Arbitrary-order double-scaling coefficients). Fix an order M . Truncate

$$- \sum_{r=1}^{M+1} \frac{a^r}{r} \left(\frac{1}{r\varepsilon} + \frac{1}{2} + \sum_{1 \leq j \leq M/2} \frac{B_{2j}}{(2j)!} (r\varepsilon)^{2j-1} \right)$$

at ε^M , substitute $a = \sum_{m=1}^{M+1} A_m \varepsilon^m$, set the constant term to $-L$, and set every positive-power coefficient through ε^M to zero. Solve in the order $A_1, A_2, \dots$. Only $r \leq M+1$ can contribute, so the computation is finite and exact.

12.3 Complex targets and logarithm branches

For a complex target $y \neq 0$, choose $L = -\text{Log } y$. Each logarithm branch produces a formal double-scaling branch. The condition $a \rightarrow 0$ selects the same local balance, but different values $L + 2\pi i k$ lead to different complex sheets. Coalescence occurs at $L = 0$, where the fixed target is $y = 1$ and the leading coefficient vanishes; a new Newton polygon is then required.

Chapter 13

Complex order near $q = 0$, resonances, and loss of identifiability

13.1 A generalized power series rather than a Taylor series

For $(a; q)_\alpha$ as in Equation (9.1), suppose $\Re\alpha > 0$. From Equation (9.2),

$$\frac{1 - q^{\alpha r}}{1 - q^r} = (1 - q^{\alpha r})(1 + q^r + q^{2r} + \cdots).$$

Unless $\alpha \in \mathbb{N}$, the expansion at $q = 0$ belongs to the generalized power semigroup generated by 1 and α , not to $\mathbb{C}[[q]]$. If $0 < \Re\alpha < 1$, the first order-dependent correction is q^α ; if $\Re\alpha > 1$, the first correction is q and is independent of α , while order dependence enters later.

Remark 13.1 (Resonance). When $\alpha = r/s$ is rational, the generalized series becomes an ordinary Puiseux series in $q^{1/s}$. When α is irrational, its support is a nonlattice additive semigroup. If two exponents coincide, coefficient resonance changes the inverse recurrence. Thus “the expansion at $q = 0$ ” for complex order cannot be stated without the arithmetic type of α .

13.2 Order identifiability collapses at zero base

For every fixed $\Re\alpha > 0$,

$$\lim_{q \rightarrow 0} (a; q)_\alpha = 1 - a.$$

Therefore the Jacobian of the map $\alpha \mapsto (a; q)_\alpha$ tends to zero. Recovering α from a noisy product value becomes singular as $q \rightarrow 0$, even though the complex-order function itself has a finite limit. This is a parameter-estimation obstruction, not a failure of analytic continuation.

13.3 Near integer order

Put $\alpha = n + \eta$. From Equation (9.3),

$$\begin{aligned} \log(a; q)_{n+\eta} &= \log(a; q)_n + \eta(\text{Log } q) \sum_{r \geq 1} \frac{a^r q^{nr}}{1 - q^r} \\ &\quad + \frac{\eta^2}{2} (\text{Log } q)^2 \sum_{r \geq 1} \frac{r a^r q^{nr}}{1 - q^r} + \mathcal{O}(\eta^3). \end{aligned} \tag{13.1}$$

This supplies a series in the order parameter itself and a local inverse $\eta(y)$. It is preferable to finite differencing integer products because it retains exact branch information through $\text{Log } q$.

Part IV

Gaussian coefficients and palindromic q -polynomials

Chapter 14

Gaussian base inversion near $q = 0$ and $q = 1$

14.1 Restricted partitions at zero base

For $0 < k < n$, let

$$G_{n,k}(q) = \left[\begin{matrix} n \\ k \end{matrix} \right]_q, \quad b = n - k, \quad d = kb.$$

Its coefficients count partitions fitting in a $k \times b$ rectangle:

$$G_{n,k}(q) = \sum_{r=0}^d p_{k,b}(r) q^r. \quad (14.1)$$

In particular $p_{k,b}(0) = p_{k,b}(1) = 1$, so the positive-base inverse at target 1 is always ordinary.

Let $u = y - 1$, and set

$$c_2 = \mathbf{1}_{k \geq 2} + \mathbf{1}_{b \geq 2}, \quad (14.2)$$

$$c_3 = \mathbf{1}_{b \geq 3} + \mathbf{1}_{k \geq 2, b \geq 2} + \mathbf{1}_{k \geq 3}. \quad (14.3)$$

These are the numbers of rectangle-compatible partitions of 2 and 3.

Theorem 14.1 (PROVED Gaussian inverse at $q = 0$). *For every $0 < k < n$,*

$$q = u - c_2 u^2 + (2c_2^2 - c_3) u^3 + \mathcal{O}(u^4). \quad (14.4)$$

More generally,

$$[u^m]q(u) = \frac{1}{m} [z^{m-1}] \left(\frac{z}{G_{n,k}(z) - 1} \right)^m.$$

Proof. Equation [Equation \(14.1\)](#) begins $u = q + c_2 q^2 + c_3 q^3 + \dots$. Apply ordinary reversion. □

For a thin rectangle, say $k = 1$, this reduces to inversion of $1 + q + \dots + q^{n-1}$. For $k, b \geq 3$, one has $c_2 = 2, c_3 = 3$, so $q = u - 2u^2 + 5u^3 + \dots$.

14.2 Bernoulli cumulants at unit base

Using

$$G_{n,k}(q) = \prod_{j=1}^k \frac{1 - q^{b+j}}{1 - q^j}$$

and $q = e^{-\varepsilon}$, define

$$\Delta_m = S_m(n) - S_m(k) - S_m(b), \quad S_m(N) = \sum_{j=1}^N j^m.$$

The elementary expansion

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} x^{2r}$$

yields

Theorem 14.2 (PROVED Complete Gaussian logarithmic jet).

$$\log G_{n,k}(e^{-\varepsilon}) = \log \binom{n}{k} - \frac{d}{2}\varepsilon + \sum_{r \geq 1} \frac{B_{2r} \Delta_{2r}}{2r(2r)!} \varepsilon^{2r}. \quad (14.5)$$

Proof. Apply the displayed scalar expansion to each numerator and denominator factor. The constant terms give $\binom{n}{k}$; the linear terms telescope to $-kb\varepsilon/2$; the even power sums give Δ_{2r} . $\square$

Put $v = \log(y/\binom{n}{k})$. Since the linear coefficient is nonzero, ordinary reversion gives an especially simple first jet.

Corollary 14.1 (PROVED Gaussian base inverse at $q = 1$). For $0 < k < n$,

$$\varepsilon = -\frac{2v}{d} + \frac{\Delta_2}{3d^3}v^2 - \frac{\Delta_2^2}{9d^5}v^3 + \mathcal{O}(v^4), \quad q = e^{-\varepsilon}. \quad (14.6)$$

The coefficient of v^4 is the first one involving Δ_4 .

Proof. In Equation (14.5), $c_1 = -d/2$, $c_2 = \Delta_2/24$, and $c_3 = 0$. Substitute these values in Equation (2.4). $\square$

Since $\Delta_2 = d(n+1)$, the variance scale that controls the centered inverse below is already visible in the quadratic coefficient.

Chapter 15

The centered-palindromic inverse theorem

15.1 A universal probabilistic normalization

Let

$$P(q) = \sum_{j=0}^D c_j q^j, \quad c_j \geq 0, \quad c_j = c_{D-j},$$

with P nonconstant. Define a random variable X by $\Pr(X = j) = c_j/P(1)$. Palindromicity is exactly the distributional symmetry $X \stackrel{d}{=} D - X$.

For real ε , set

$$H_P(\varepsilon) = \frac{e^{D\varepsilon/2} P(e^{-\varepsilon})}{P(1)} = \mathbb{E} e^{-\varepsilon(X-D/2)}. \quad (15.1)$$

Theorem 15.1 (PROVED Global two-branch inverse). *The function H_P is even, satisfies $H_P(0) = 1$, is strictly increasing on $(0, \infty)$, and tends to infinity there. Hence for every $h > 1$ the equation $H_P(\varepsilon) = h$ has exactly two real solutions $\varepsilon = \pm E_P(h)$. The inverse has a square-root branch point at $h = 1$.*

Proof. Let $Z = X - D/2$. Symmetry gives $Z \stackrel{d}{=} -Z$, hence

$$H_P(\varepsilon) = \mathbb{E} \cosh(\varepsilon Z).$$

It is even and at least one. For $\varepsilon > 0$,

$$H'_P(\varepsilon) = \mathbb{E}[Z \sinh(\varepsilon Z)] > 0,$$

because $z \sinh(\varepsilon z) \geq 0$ and the nonconstant polynomial makes it positive with nonzero probability. Since $c_0 = c_D > 0$, the endpoint terms force exponential growth. Finally, $H_P(\varepsilon) = 1 + \text{Var}(X)\varepsilon^2/2 + \dots$, so the local inverse is square-root. $\square$

This theorem resolves a common apparent paradox. The unnormalized Gaussian polynomial has an ordinary inverse at $q = 1$ because of its deterministic linear drift $D/2$. Removing that drift exposes the reciprocal symmetry and turns $q = 1$ into a genuine two-sheeted branch point.

15.2 Cumulant Puiseux series

Let κ_{2r} be the even cumulants of Z ; odd cumulants vanish. Put $u = \log H_P(\varepsilon)$. Then

$$u = \frac{\kappa_2}{2!} \varepsilon^2 + \frac{\kappa_4}{4!} \varepsilon^4 + \frac{\kappa_6}{6!} \varepsilon^6 + \dots$$

Reversion in ε^2 gives

Theorem 15.2 (**PROVED** Centered inverse through order $u^{5/2}$).

$$\varepsilon = \pm \sqrt{\frac{2u}{\kappa_2}} \left[1 - \frac{\kappa_4}{12\kappa_2^2} u + \left(\frac{7\kappa_4^2}{288\kappa_2^4} - \frac{\kappa_6}{180\kappa_2^3} \right) u^2 + \mathcal{O}(u^3) \right]. \quad (15.2)$$

Proof. Write $u = A\varepsilon^2 + B\varepsilon^4 + C\varepsilon^6 + \dots$ and substitute $\varepsilon = (u/A)^{1/2}(1 + \alpha u + \beta u^2 + \dots)$. Solving the coefficients gives $\alpha = -B/(2A^2)$ and $\beta = (-4AC + 7B^2)/(8A^4)$. Insert $A = \kappa_2/2$, $B = \kappa_4/24$, $C = \kappa_6/720$. $\square$

15.3 Gaussian cumulants

For $P = G_{n,k}$, comparison with Equation (14.5) gives

$$\kappa_{2r} = \frac{B_{2r}}{2r} \Delta_{2r}, \quad \kappa_2 = \frac{k(n-k)(n+1)}{12}. \quad (15.3)$$

Thus Equation (15.2) is an explicit Bernoulli–Faulhaber expansion. The same construction applies to any positive palindromic q -enumerator; no product representation is required.

Chapter 16

Roots of unity and the square-free Gaussian covering

16.1 Cyclotomic factorization

Let $\Phi_m(q)$ be the m -th cyclotomic polynomial. Counting multiples of m in the product formula yields

$$G_{n,k}(q) = \prod_{m=2}^n \Phi_m(q)^{e_m}, \quad e_m = \left\lfloor \frac{n}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{n-k}{m} \right\rfloor. \quad (16.1)$$

Theorem 16.1 (PROVED Gaussian polynomials are square-free). *For $0 < k < n$, every exponent e_m in Equation (16.1) is either 0 or 1. Consequently every complex zero of $G_{n,k}$ is a simple root of unity.*

Proof. Write $n = k + b$. Then

$$e_m = \left\lfloor \frac{k+b}{m} \right\rfloor - \left\lfloor \frac{k}{m} \right\rfloor - \left\lfloor \frac{b}{m} \right\rfloor.$$

The difference records whether the two residues modulo m produce a carry, and hence is 0 or 1. Cyclotomic polynomials are separable in characteristic zero and mutually coprime. $\square$

If ζ is primitive of order m and $e_m = 1$, write

$$R_m(q) = \prod_{r \neq m} \Phi_r(q)^{e_r}.$$

Then

$$G'_{n,k}(\zeta) = \Phi'_m(\zeta) R_m(\zeta) \neq 0, \quad q(y) = \zeta + \frac{y}{\Phi'_m(\zeta) R_m(\zeta)} + \mathcal{O}(y^2). \quad (16.2)$$

Thus Gaussian root-of-unity inverses are ordinary. Higher Puiseux behavior at roots of unity belongs to q -multinomials and q -factorials, not to Gaussian polynomials themselves.

16.2 The alternating base

At $q = -1$,

$$G_{n,k}(-1) = \begin{cases} 0, & n \text{ even and } k \text{ odd,} \\ \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \text{otherwise.} \end{cases} \quad (16.3)$$

The zero case is always simple because $e_2 = 1$.

Theorem 16.2 (**PROVED** Exact derivative of the alternating zero). *If $n = 2m$ and $k = 2r + 1$, with $0 \leq r < m$, then*

$$G'_{2m,2r+1}(-1) = (m - r) \binom{m}{r}. \quad (16.4)$$

Hence

$$q = -1 + \frac{y}{(m - r) \binom{m}{r}} + \mathcal{O}(y^2)$$

for the inverse branch through the alternating zero.

Proof. In the product

$$G_{2m,2r+1}(q) = \prod_{j=1}^{2r+1} \frac{1 - q^{2(m-r)-1+j}}{1 - q^j},$$

there is one more even numerator exponent than even denominator exponent. For an even exponent $2s$, $(1 - q^{2s})/(1 + q) \rightarrow 2s$; every odd factor tends to 2. The ratio of the even leading coefficients is $2m!/(r!(m - r - 1)!)$, while the odd factors contribute $1/2$. Their product is $(m - r) \binom{m}{r}$. $\square$

16.3 Nonzero targets at a root of unity

Even when $G(\zeta) \neq 0$, the derivative may vanish and create a fold at a nonzero target. These points are zeros of $G'(q)$ constrained to the root-of-unity locus. Their elimination ideal is

$$\langle \Phi_m(q), G'(q) \rangle.$$

For fixed (n, k, m) it is decidable by exact resultants. A systematic table of such critical coincidences is a natural continuation of the earlier secondary-discriminant computations.

Chapter 17

Large base and reciprocal Gaussian branches

17.1 Reciprocity

The rectangle complement involution gives

$$G_{n,k}(q) = q^d G_{n,k}(q^{-1}), \quad d = k(n - k). \quad (17.1)$$

Let

$$G_{n,k}(Q) = 1 + Q + c_2 Q^2 + c_3 Q^3 + \cdots$$

with c_2, c_3 as in Equation (14.3), and put $Y = y^{1/d}$ on a chosen branch.

Theorem 17.1 (PROVED Large-target Gaussian inverse).

$$q = Y - \frac{1}{d} + \left(\frac{d-1}{2d^2} - \frac{c_2}{d} \right) Y^{-1} + \mathcal{O}(Y^{-2}). \quad (17.2)$$

The coefficient of Y^{-2} is

$$-\frac{(d-1)(2d-1)}{3d^3} + \frac{2c_2(d-1)}{d^2} - \frac{c_3}{d}.$$

Proof. Substitute $q = Y(1 + A_1 Y^{-1} + A_2 Y^{-2} + \cdots)$ into $q^d G(q^{-1}) = y$ and compare powers of Y^{-1} . The equations are triangular and give the displayed coefficients. $\square$

For $q \rightarrow -\infty$, reciprocity remains valid. If d is odd, the leading real target tends to $-\infty$; if d is even, it tends to $+\infty$. The complex branches are indexed by the d -th roots in Y .

Chapter 18

Continuous upper and lower parameters; q-multinomials

18.1 q-gamma interpolation of the Gaussian coefficient

For $0 < q < 1$, define

$$\mathcal{G}_q(\alpha, \beta) = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}. \quad (18.1)$$

It agrees with $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at integers. Let ψ_q denote the q -digamma function. Then

$$\partial_\alpha \log \mathcal{G}_q = \psi_q(\alpha + 1) - \psi_q(\alpha - \beta + 1), \quad (18.2)$$

$$\partial_\beta \log \mathcal{G}_q = -\psi_q(\beta + 1) + \psi_q(\alpha - \beta + 1). \quad (18.3)$$

Theorem 18.1 (PROVED Parameter branch geometry). *Assume $\alpha > \beta > 0$.*

1. *Since ψ_q is strictly increasing, $\mathcal{G}_q$ is strictly increasing in α . Its upper-parameter inverse is therefore unique on every interval contained in this domain.*
2. *As a function of $\beta \in (0, \alpha)$, it is symmetric about $\alpha/2$, strictly increasing before the midpoint and strictly decreasing after it. At the midpoint,*

$$\partial_\beta^2 \log \mathcal{G}_q = -2\psi'_q(\alpha/2 + 1) < 0.$$

Thus the lower-parameter inverse has two real branches below the maximum and a square-root merger at the maximum.

Proof. For $0 < q < 1$, $\psi'_q(x) > 0$. The signs follow directly from [Equations \(18.2\) and \(18.3\)](#); differentiating the second identity proves the midpoint statement. $\square$

Near $q = 1$, the q -gamma expansion derived later implies

$$\log \mathcal{G}_{e^{-\varepsilon}}(\alpha, \beta) = \log \frac{\Gamma(\alpha + 1)}{\Gamma(\beta + 1)\Gamma(\alpha - \beta + 1)} - \frac{\beta(\alpha - \beta)}{2}\varepsilon + \mathcal{O}(\varepsilon^2). \quad (18.4)$$

Hence the base inverse is ordinary whenever $\beta(\alpha - \beta) \neq 0$, and mixed inverses in (q, α, β) follow from the Jacobian formulas of Part I.

18.2 q-multinomial cyclotomic carry theorem

For nonnegative integers $k_1, \dots, k_r$ with $n = \sum_i k_i$, write

$$\left[\begin{matrix} n \\ k_1, \dots, k_r \end{matrix} \right]_q = \frac{(q; q)_n}{\prod_{i=1}^r (q; q)_{k_i}}.$$

At a primitive d -th root, the cyclotomic exponent is

$$e_d = \left\lfloor \frac{n}{d} \right\rfloor - \sum_{i=1}^r \left\lfloor \frac{k_i}{d} \right\rfloor. \quad (18.5)$$

Theorem 18.2 (PROVED Residue-carry ramification). *Let $r_i \equiv k_i \pmod{d}$, $0 \leq r_i < d$. Then*

$$e_d = \left\lfloor \frac{r_1 + \dots + r_r}{d} \right\rfloor, \quad 0 \leq e_d \leq r - 1. \quad (18.6)$$

If $e_d = m > 0$, the q -multinomial has a zero of order m at every primitive d -th root and its base inverse at target zero has m Puiseux branches.

Proof. Write $k_i = d\ell_i + r_i$. Since $n = d\sum_i \ell_i + \sum_i r_i$, substitution in Equation (18.5) gives Equation (18.6). The cyclotomic factor $\Phi_d^{e_d}$ has that exact zero order, and Newton–Puiseux gives the inverse branches. $\square$

For $r = 2$, the carry count is at most one, recovering Gaussian square-freeness. For many parts it can be arbitrarily large. This is the simplest exact mechanism by which passing from q -binomials to another standard q -analog changes an ordinary root-of-unity inverse into a higher Puiseux inverse.

18.3 Centered q-multinomial inverse

The q -multinomial has nonnegative palindromic coefficients and degree

$$D = \sum_{1 \leq i < j \leq r} k_i k_j.$$

Therefore Equation (15.1) applies verbatim. Its centered base inverse has exactly two global real branches, and its local square-root coefficients are the cumulants of the multiset-inversion statistic. In contrast, its root-of-unity inverse can have arbitrarily high finite ramification by Equation (18.6); the same polynomial thus exhibits two different singular mechanisms at different targets.

Part V

Other principal q -analogs

Chapter 19

q-numbers and q-factorials

19.1 The q-number: one exact inverse and one algebraic inverse

For $q \neq 1$, define

$$[x]_q = \frac{1 - q^x}{1 - q}, \quad q^x = e^{x \operatorname{Log} q}.$$

Inversion in x is elementary once a logarithm branch is chosen:

$$x = \frac{\operatorname{Log}(1 - (1 - q)y)}{\operatorname{Log} q}. \quad (19.1)$$

The branch points are inherited from both logarithms. When $x = m \in \mathbb{N}$, inversion in q is algebraic because

$$[m]_q = 1 + q + \cdots + q^{m-1}$$

.

At $q = 0$, let $u = y - 1$. Then

$$\begin{aligned} m = 2 : \quad q &= u, & \text{exactly,} \\ m = 3 : \quad q &= u - u^2 + 2u^3 + \mathcal{O}(u^4), \\ m \geq 4 : \quad q &= u - u^2 + u^3 + \mathcal{O}(u^4). \end{aligned}$$

At $q = e^{-\varepsilon} \rightarrow 1$,

$$\begin{aligned} [x]_{e^{-\varepsilon}} &= x + \frac{x(1-x)}{2}\varepsilon + \frac{x(2x^2-3x+1)}{12}\varepsilon^2 \\ &\quad - \frac{x^2(x-1)^2}{24}\varepsilon^3 + \mathcal{O}(\varepsilon^4). \end{aligned} \quad (19.2)$$

Thus, for $x \notin \{0, 1\}$,

$$\varepsilon = \frac{2(y-x)}{x(1-x)} + \frac{2(2x-1)}{3x^2(x-1)^2}(y-x)^2 + \mathcal{O}((y-x)^3). \quad (19.3)$$

The values $x = 0$ and $x = 1$ are unidentifiable because the q-number is identically 0 or 1.

For integer $m \geq 2$, the alternating base is regular:

$$[2r]_{-1} = 0, \quad \partial_q[2r]_q|_{q=-1} = r; \quad [2r+1]_{-1} = 1, \quad \partial_q[2r+1]_q|_{q=-1} = -r.$$

At infinity, $[m]_q = q^{m-1}(1 + q^{-1} + \cdots)$, so the inverse has $m - 1$ Puiseux branches.

19.2 q-factorials at the distinguished bases

Let

$$[n]_q! = \prod_{j=1}^n [j]_q = \Gamma_q(n+1).$$

At zero base,

$$[n]_q! = 1 + (n-1)q + \frac{(n-2)(n+1)}{2}q^2 + \mathcal{O}(q^3). \quad (19.4)$$

For $n \geq 2$ and $u = y - 1$,

$$q = \frac{u}{n-1} - \frac{(n-2)(n+1)}{2(n-1)^3}u^2 + \mathcal{O}(u^3). \quad (19.5)$$

Theorem 19.1 (PROVED Exact alternating ramification of q-factorials). *Let $m = \lfloor n/2 \rfloor$. Then*

$$[n]_q! = m! (q+1)^m (1 + \mathcal{O}(q+1)) \quad (q \rightarrow -1). \quad (19.6)$$

Hence the inverse at target zero is

$$q = -1 + \omega_m^j \left(\frac{y}{m!} \right)^{1/m} + \mathcal{O}(y^{2/m}), \quad 0 \leq j < m. \quad (19.7)$$

Proof. Exactly the even q-numbers vanish. For $j = 2r$, $[2r]_q = r(q+1) + \mathcal{O}((q+1)^2)$, while every odd q-number tends to 1. Multiplying $r = 1, \dots, m$ proves the result. $\square$

More generally, at a primitive d -th root ζ ,

$$\text{ord}_{q=\zeta} [n]_q! = \left\lfloor \frac{n}{d} \right\rfloor, \quad d > 1. \quad (19.8)$$

This follows because $[j]_q$ has a simple zero exactly when $d \mid j$. Thus q-factorials supply arbitrarily high root-of-unity inverse ramification.

At infinity, with $N = \binom{n}{2}$,

$$[n]_q! = q^N \left(1 + \frac{n-1}{q} + \mathcal{O}(q^{-2}) \right),$$

so, on a chosen root branch,

$$q = y^{1/N} - \frac{n-1}{N} + \mathcal{O}(y^{-1/N}) = y^{1/N} - \frac{2}{n} + \mathcal{O}(y^{-1/N}). \quad (19.9)$$

Chapter 20

q-gamma and q-digamma inverse jets

20.1 Definitions and parameter derivatives

For $0 < q < 1$,

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}. \quad (20.1)$$

It obeys $\Gamma_q(x+1) = [x]_q \Gamma_q(x)$ and $\Gamma_q(1) = \Gamma_q(2) = 1$. The logarithmic derivative is

$$\psi_q(x) = -\log(1-q) + (\log q) \sum_{j \geq 0} \frac{q^{x+j}}{1-q^{x+j}}, \quad (20.2)$$

and

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1-q^{x+j})^2} > 0. \quad (20.3)$$

Thus Γ_q has the same two-branch real argument geometry as the ordinary gamma function: its logarithm is strictly convex and has one minimum.

20.2 All-order algebraic expansion at $q = 1$

Set $q = e^{-\varepsilon}$. The following coefficient functions are polynomials in x :

$$c_1(x) = -\frac{(x-1)(x-2)}{4}, \quad (20.4)$$

$$c_{2r}(x) = \frac{B_{2r}}{2r(2r)!} \left(\frac{B_{2r+1}(x) - B_{2r+1}(1)}{2r+1} - (x-1) \right), \quad r \geq 1. \quad (20.5)$$

Theorem 20.1 (PROVED q-gamma expansion and odd-power cancellation). *Uniformly for x in compact subsets of the positive half-plane,*

$$\log \Gamma_{e^{-\varepsilon}}(x) \sim \log \Gamma(x) + c_1(x)\varepsilon + \sum_{r \geq 1} c_{2r}(x)\varepsilon^{2r}. \quad (20.6)$$

No odd algebraic power beyond ε^1 occurs. In particular,

$$c_2(x) = \frac{(x-2)(x-1)(2x+3)}{144}, \quad (20.7)$$

$$c_4(x) = -\frac{(x-2)(x-1)(6x^3 + 3x^2 + 7x + 15)}{86400}. \quad (20.8)$$

Proof. Subtract $\log \Gamma(x)$ from the logarithm of the functional equation. By Equation (19.2),

$$\log \frac{[x]e^{-\varepsilon}}{x} = -\frac{x-1}{2}\varepsilon + \sum_{r \geq 1} \frac{B_{2r}(x^{2r}-1)}{2r(2r)!} \varepsilon^{2r}. \quad (20.9)$$

If $C_m(x)$ is the coefficient of ε^m in the left side of Equation (20.6), then

$$C_m(x+1) - C_m(x) = [\varepsilon^m] \log \frac{[x]e^{-\varepsilon}}{x}, \quad C_m(1) = 0.$$

The Bernoulli identity $B_{r+1}(x+1) - B_{r+1}(x) = (r+1)x^r$ gives Equations (20.4) and (20.5). The product asymptotics of Moak and McIntosh [10, 9] select these polynomial solutions and justify the asymptotic expansion; the difference source has no odd powers beyond one, so neither does the selected solution. $\square$

20.3 Inverting the base

Fix $x \notin \{1, 2\}$, and put

$$v = \log \frac{y}{\Gamma(x)}.$$

Since $c_1(x) \neq 0$, the branch $q \rightarrow 1$ is ordinary:

$$\varepsilon = \frac{v}{c_1} - \frac{c_2}{c_1^3} v^2 + \frac{2c_2^2}{c_1^5} v^3 + \mathcal{O}(v^4), \quad q = e^{-\varepsilon}. \quad (20.10)$$

At $x = 1$ and $x = 2$, $\Gamma_q(x) \equiv 1$, so no local base inverse exists. These are complete, not merely first-order, degeneracies.

20.4 Inverting the argument

Let x_0 be a chosen branch of the ordinary inverse gamma function, $\Gamma(x_0) = y$, and assume $\psi(x_0) \neq 0$. Seek $x = x_0 + A\varepsilon + B\varepsilon^2 + \dots$. From Equation (20.6),

$$A = -\frac{c_1(x_0)}{\psi(x_0)}, \quad (20.11)$$

$$B = -\frac{\frac{1}{2}\psi'(x_0)A^2 + c'_1(x_0)A + c_2(x_0)}{\psi(x_0)}. \quad (20.12)$$

At the unique positive minimum, $\psi(x_0) = 0$ and $\psi'(x_0) > 0$; the two inverse-gamma branches merge with square-root scaling. A simultaneous perturbation of the target and q is handled by the fold-unfolding model of Part I.

20.5 q-digamma argument and base inverses

Equation (20.3) shows that $x \mapsto \psi_q(x)$ is strictly increasing. Its range is $(-\infty, -\log(1-q))$, so every target in that interval has exactly one positive real argument inverse.

Differentiating Equation (20.6) gives

$$\psi_{e^{-\varepsilon}}(x) \sim \psi(x) + c'_1(x)\varepsilon + \sum_{r \geq 1} c'_{2r}(x)\varepsilon^{2r}. \quad (20.13)$$

For a fixed target and an ordinary inverse-digamma point x_0 ,

$$x = x_0 - \frac{c'_1(x_0)}{\psi'(x_0)}\varepsilon + \mathcal{O}(\varepsilon^2).$$

For base inversion at fixed x , the linear coefficient

$$c'_1(x) = \frac{3 - 2x}{4}$$

vanishes only at $x = 3/2$. Since $c'_2(3/2) = -1/288$, this point has a square-root base inverse:

$$\psi_{e^{-\varepsilon}}(3/2) - \psi(3/2) = -\frac{\varepsilon^2}{288} + \mathcal{O}(\varepsilon^4). \quad (20.14)$$

All other fixed positive x have an ordinary base inverse at $q = 1$.

Chapter 21

q-beta: coordinate inverses and a critical base curve

21.1 Coordinate monotonicity

Define

$$B_q(x, y) = \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}, \quad x, y > 0. \quad (21.1)$$

Then

$$\partial_x \log B_q(x, y) = \psi_q(x) - \psi_q(x+y) < 0, \quad (21.2)$$

$$\partial_y \log B_q(x, y) = \psi_q(y) - \psi_q(x+y) < 0. \quad (21.3)$$

Hence each coordinate inverse is unique on its positive real domain whenever the target lies in the corresponding range. Its local coefficients follow by differentiating Equation (21.3); the first derivative is the reciprocal of the displayed difference.

21.2 Base expansion and the curve $xy = 1$

Combining three copies of Equation (20.6) gives

$$\log \frac{B_{e^{-\varepsilon}}(x, y)}{B(x, y)} = b_1(x, y)\varepsilon + b_2(x, y)\varepsilon^2 + \mathcal{O}(\varepsilon^4), \quad (21.4)$$

where

$$b_1(x, y) = \frac{xy - 1}{2}, \quad (21.5)$$

$$b_2(x, y) = -\frac{x^2y + xy^2 - xy - 1}{24}. \quad (21.6)$$

There is no cubic term.

Theorem 21.1 (PROVED Type change of the q-beta base inverse). *At $q = 1$, the inverse in q is ordinary if $xy \neq 1$. On the curve $xy = 1$,*

$$b_2(x, 1/x) = -\frac{(x-1)^2}{24x}. \quad (21.7)$$

Thus every positive point of that curve except $(1, 1)$ is a quadratic fold and has a square-root base inverse. At $(1, 1)$, $B_q(1, 1) \equiv 1$, so the base is completely unidentifiable.

Proof. Off $xy = 1$, Equation (21.5) and the inverse-function theorem apply. On the curve substitute $y = 1/x$ into Equation (21.6). The result is nonzero unless $x = 1$. The identity at $(1, 1)$ follows from $\Gamma_q(1) = \Gamma_q(2) = 1$. $\square$

If $v = \log(B_q/B)$, then off the critical curve

$$\varepsilon = \frac{v}{b_1} - \frac{b_2}{b_1^3}v^2 + \mathcal{O}(v^3).$$

On $xy = 1$, $x \neq 1$,

$$\varepsilon = \pm \sqrt{\frac{v}{b_2}} + \mathcal{O}(v).$$

For the real branch $0 < q < 1$, retain the sign with $\varepsilon > 0$.

Chapter 22

Euler q-exponentials, basic hypergeometric series, and theta products

22.1 Euler q-exponentials reduce exactly to Pochhammer inversion

For $0 < |q| < 1$, use the conventions

$$e_q(z) = \frac{1}{(z; q)_\infty}, \quad E_q(z) = (-z; q)_\infty.$$

If $\mathcal{A}_q$ denotes a selected inverse branch of the infinite product in its argument, then

$$e_q^{-1}(y) = \mathcal{A}_q(y^{-1}), \quad E_q^{-1}(y) = -\mathcal{A}_q(y). \quad (22.1)$$

These identities transport every zero, critical point, endpoint expansion, and branch label from Part III without recomputation. Base inversion of the q-exponentials is likewise the base inversion of the corresponding product with a transformed target.

Near $z = 0$, [Equation \(10.2\)](#) immediately gives

$$e_q^{-1}(1 + s) = (1 - q)s + \mathcal{O}(s^2), \quad E_q^{-1}(1 + s) = (1 - q)s + \mathcal{O}(s^2),$$

with different higher coefficients because one target is reciprocated and the other argument is negated.

22.2 A coefficientwise inverse for basic hypergeometric series

Let a chosen basic hypergeometric function be written locally as

$$F(z; \mathbf{a}, \mathbf{b}, q) = 1 + c_1 z + c_2 z^2 + c_3 z^3 + \cdots, \quad c_1 \neq 0.$$

Then the argument inverse is universally

$$z = \frac{y - 1}{c_1} - \frac{c_2}{c_1^3}(y - 1)^2 + \left(\frac{2c_2^2}{c_1^5} - \frac{c_3}{c_1^4} \right) (y - 1)^3 + \cdots. \quad (22.2)$$

Every c_n is a quotient of finite q-Pochhammer products, so its derivatives in any numerator or denominator parameter reduce to finite sums of the form [Equation \(5.2\)](#). This gives mixed inverse jets in the argument, base, and all shape parameters.

If a numerator parameter crosses $a_i = q^{-N}$, the series truncates. The argument inverse then becomes an algebraic covering of degree at most N . If simultaneously a denominator parameter approaches a cancelling value, one must first cancel common q-Pochhammer factors; otherwise a removable singularity is incorrectly reported as an inverse branch point.

22.3 Parameter inversion near a truncation locus

Write $a_i = q^{-N}(1 + \eta)$. The factor $(a_i; q)_n$ contains one small term when $n > N$, so the tail of the hypergeometric series is $\mathcal{O}(\eta)$. At $\eta = 0$, the function is a polynomial $F_N(z)$. If $\partial_\eta F \neq 0$, η has an ordinary inverse at fixed z . If that derivative vanishes at a root or critical point of F_N , the joint Newton polygon in $(z - z_0, \eta, y - y_0)$ determines the fractional scaling. This is the basic-hypergeometric analogue of the finite-product fold unfolding.

22.4 Multiplicative theta products

Use

$$\vartheta(z; q) = (z; q)_\infty (q/z; q)_\infty (q; q)_\infty. \quad (22.3)$$

For fixed $0 < |q| < 1$, its zeros $z = q^m$, $m \in \mathbb{Z}$, are simple. Hence the argument inverse at target zero is ordinary on every sheet:

$$z = q^m + \frac{y}{\partial_z \vartheta(q^m; q)} + \mathcal{O}(y^2).$$

The quasi-periodicity $\vartheta(qz; q) = -z^{-1} \vartheta(z; q)$ transports the derivative normalization from one zero to all others.

Base inversion is more delicate. Near $q = 0$, the Laurent/Fourier series gives an ordinary or Puiseux jet according to the first nonzero q-power. Near $q = 1$, modular transformation produces exponential scales $e^{-\pi^2/\varepsilon}$, so a complete inverse is a transseries rather than a Bernoulli power series. The dominant exponential is inverted by a logarithm or Lambert W ; the modular images determine the Stokes corrections.

22.5 Other positive palindromic q-enumerators

Whenever a finite q-analog is a nonconstant polynomial with nonnegative palindromic coefficients, the centered theorem of Part IV applies without change. Examples include Poincare polynomials of finite Coxeter groups, q-multinomials, many Gaussian quotients, and selected q-Catalan and plane partition enumerators under their palindromic normalizations. Objects lacking palindromicity still have ordinary local inverse jets, but lose the exact reciprocal two-branch geometry.

Part VI

Fabius–Rvachev synthesis, algorithms, and frontiers

Chapter 23

An effective geometric base for the Rvachev infinite sinc product

23.1 A one-parameter compactly supported convolution family

Let $\text{sinc } z = \sin z/z$, and for $0 < q < 1$ define

$$\widehat{U}_q(t) = \prod_{j \geq 0} \text{sinc}((1-q)q^j t). \quad (23.1)$$

Probabilistically, this is the characteristic function of the sum of independent centered uniform random variables whose half-widths are $(1-q)q^j$. Their widths sum to one, so the corresponding density is supported on $[-1, 1]$. At $q = 1/2$,

$$\widehat{U}_{1/2}(t) = \prod_{r \geq 1} \text{sinc}(t/2^r),$$

which is the standard infinite sinc product underlying the Rvachev up-function, up to Fourier-transform normalization. Its cumulative form is correspondingly linked to the Fabius function in the conventions developed by the source corpus.

23.2 Cumulants and a globally identifiable base

Euler's sine product gives

$$\log \text{sinc } z = - \sum_{m \geq 1} \frac{\zeta(2m)}{m\pi^{2m}} z^{2m}.$$

Therefore

$$\log \widehat{U}_q(t) = - \sum_{m \geq 1} \frac{\zeta(2m)}{m\pi^{2m}} R_m(q) t^{2m}, \quad R_m(q) = \frac{(1-q)^{2m}}{1-q^{2m}}. \quad (23.2)$$

Theorem 23.1 (**PROVED** Every even cumulant identifies the geometric base). *For every $m \geq 1$, R_m is strictly decreasing from 1 to 0 on $(0, 1)$. Hence any one nonzero even cumulant coefficient in Equation (23.2) determines q uniquely.*

Proof. Differentiate the logarithm:

$$\frac{R'_m(q)}{R_m(q)} = -\frac{2m}{1-q} + \frac{2mq^{2m-1}}{1-q^{2m}}.$$

The second fraction is smaller than the first because $q^{2m-1}(1-q) < 1 - q^{2m}$, equivalently $q^{2m-1} < 1$. The endpoint limits are immediate. $\square$

At the dyadic point,

$$R_m(1/2) = \frac{1}{2^{2m} - 1}, \quad R'_m(1/2) = -\frac{4m(2^{2m} - 2)}{(2^{2m} - 1)^2}. \quad (23.3)$$

Thus if an observed normalized cumulant ratio is $R_m = 1/(2^{2m} - 1) + \delta$, its effective-base inverse is

$$q = \frac{1}{2} - \frac{(2^{2m} - 1)^2}{4m(2^{2m} - 2)}\delta + \mathcal{O}(\delta^2). \quad (23.4)$$

This gives a direct inverse bridge between Fabius–Rvachev moment data and the q-analog machinery of the preceding parts.

23.3 Overdetermined consistency and deformations

A genuine geometric convolution must return the same q from every even cumulant. Disagreement between the inverses of $R_1, R_2, \dots$ therefore measures departure from exact geometric scaling. With an additional global scale s , replace each width by $s(1 - q)q^j$; the $2m$ -th logarithmic coefficient becomes $s^{2m}R_m(q)$. Two independent cumulants generically recover (s, q) , while further cumulants test the model.

The same idea applies to finite geometric combs. Their Fourier products are finite q-products in the squared frequency, and their interpolation weights contain Gaussian coefficients. Inverting an observed comb response for its ratio q is therefore a simultaneous inverse problem in the sense of [Equation \(4.1\)](#), with the dyadic comb $q = 1/2$ as an ordinary interior base point rather than a singular endpoint.

Chapter 24

Numerical realization of the inverse atlas

24.1 Coordinate selection before iteration

A reliable solver first changes coordinates so that the desired branch is regular:

Regime	Forward coordinate	Inverse coordinate
$q \approx 0$	q or $q^{1/s}$	ordinary/Puiseux series
$q \approx 1$	$\varepsilon = -\text{Log } q$	logarithmic target $w = \text{Log}(y/y_0)$
$q \approx -1$	$h = q + 1$	$h = (y - y_0)^{1/m}v$
root ζ	$h = q - \zeta$	cyclotomic multiplicity coordinate
$q \approx \infty$	$Q = q^{-1}$	$Y = y^{1/D}$, then Laurent reversion
fixed-target infinite product	(a, ε)	weighted degree $a \sim \varepsilon$

Using Newton iteration in the original variable near a fold loses half the digits and can jump sheets. Iteration in the desingularized coordinate is both better conditioned and branch explicit.

24.2 Predictor–corrector continuation

For a regular branch $F(p, \lambda) = y$, differentiate along a continuation path $(y(s), \lambda(s))$:

$$\frac{dp}{ds} = \frac{y'(s) - F_\lambda \lambda'(s)}{F_p}. \quad (24.1)$$

An Euler or higher-order predictor from [Equation \(24.1\)](#) is followed by a Newton corrector. Near $F_p = 0$, switch to pseudo-arclength continuation of the curve (p, y) ; this traverses a fold without choosing an artificial vertical graph. A branch label consists of the starting germ plus the ordered list of discriminant cuts crossed.

24.3 Exact root-of-unity preprocessing

For polynomial q -analogs, factor cyclotomically before numerical work.

1. Compute the exponent of every Φ_d by floor formulas rather than by floating-point factorization.
2. Divide out the exact cyclotomic factor and evaluate the nonzero cofactor at ζ .
3. Use the multiplicity as the Puiseux denominator and the cofactor to normalize the leading root.

This prevents a cluster of nearly coincident numerical roots from being mistaken for distinct simple branches.

24.4 Certified local inverses

Suppose a truncated inverse jet $p_M(y)$ is available on a disk. Compute the residual

$$R_M(y) = F(p_M(y)) - y.$$

If an interval or ball enclosure proves F_p stays away from zero, then a Newton–Kantorovich bound turns R_M into an error bound for the true branch. At a ramification point, apply the same argument to the desingularized equation in $v = (p - p_0)/(y - y_0)^{1/m}$. Rouché’s theorem can additionally certify the number of branches inside a contour.

24.5 Reproducible experiments in the archive

The accompanying program `inverse_q_analog_experiments.py` uses exact SymPy algebra and high-precision mpmath arithmetic. It verifies the following independent claims.

1. The parameter [Equation \(7.3\)](#), the logarithmic second derivatives [Equations \(7.4\)](#) and [\(7.5\)](#), and the exact cubic [Equation \(7.6\)](#).
2. Cyclotomic carry counts, Gaussian square-freeness in sample families, the q-factorial coefficient $m!$ at $q = -1$, and [Equation \(16.4\)](#).
3. The five-term double-scaling inverse [Equation \(12.1\)](#) at two targets and three decreasing values of ε .
4. The q-gamma coefficients [Equations \(20.4\)](#), [\(20.7\)](#) and [\(20.8\)](#) and the absence of a cubic algebraic term.
5. The centered Gaussian Puiseux expansion for $(n, k) = (8, 3)$, where

$$\kappa_2 = 15, \quad \kappa_4 = -111, \quad \kappa_6 = 2520.$$

6. The q-beta coefficients and the reduction $b_2(x, 1/x) = -(x - 1)^2/(24x)$.

A representative double-scaling check is

y	ε	numerical a	five-term error
0.3	0.05	0.0560243496070	-8.05×10^{-9}
0.3	0.025	0.0288380205105	-1.29×10^{-10}
0.7	0.05	0.0173299163772	-4.24×10^{-11}
0.7	0.025	0.00878272387187	-6.70×10^{-13}

The archive includes the full text output so that no rounded table is the sole record of a computation.

Chapter 25

Conjectures and research directions

25.1 Algebraic monodromy of finite base inverses

For fixed generic (a, y) , the equation $P_n(a, q) = y$ has degree $N = \binom{n}{2}$ in q . Its coefficients are highly structured but possess no obvious decomposition for generic parameters.

Conjecture 25.1 (Generic symmetric monodromy). For $n \geq 3$, after removing the obvious degenerate loci $a \in \{0, 1\}$, y critical, and lower-degree specializations, the geometric monodromy group of $P_n(a, q) - y$ over the (a, y) -parameter space is the full symmetric group S_N .

A computational route is to specialize to several rational pairs (a, y) , compute exact Galois groups modulo good primes, and then prove that a transposition and a long cycle occur in the generic monodromy. The explicit folds at $q = -1$ naturally supply transposition candidates.

25.2 Secondary discriminants and collision normal forms

The primary discriminant eliminates q from $P_n - y = 0$ and $\partial_q P_n = 0$. A secondary discriminant detects equality of two distinct critical values. The first inverse report found a positive-base collision for $(a; q)_5$. The next objective is a stratified classification rather than a list of examples.

Conjecture 25.2 (Generic normal crossings of critical-value collisions). Away from roots of unity, identically constant parameters, and higher critical points, the secondary-discriminant hypersurface for finite q -Pochhammer base inversion is reduced, and a generic point represents the transverse equality of exactly two simple critical values.

A proof would combine elimination theory with transversality of the critical value map. Exceptional strata should correspond to the vanishing of explicit higher resultants.

25.3 Resurgence of $q \rightarrow 1$ inverse jets

The Bernoulli coefficients in [Equation \(11.2\)](#) grow factorially, and modular/theta analogs contain exponentially small scales invisible to the power series.

Conjecture 25.3 (Sectorial Borel summability). For fixed a away from the polylogarithmic branch locus, the formal base inverse [Equation \(11.3\)](#) and the corresponding q -gamma and q -beta inverse jets are Gevrey-1 and Borel summable in every non-Stokes sector. Their Stokes discontinuities are generated by the nearest dilogarithmic or modular saddle actions.

The technically promising route is to begin from integral representations of q -shifted factorials, prove uniform alien-derivative estimates for the forward transseries, and use closure of resurgent functions under local composition and reversion.

25.4 Global q -beta critical surfaces

The local unit-base transition is the simple curve $xy = 1$, but numerical sampling shows that the sign of $\partial_q B_q(x, y)$ can change inside $(0, 1)$ for several positive parameter pairs. Thus the global critical set is richer than a continuation of $xy = 1$.

Research problem 25.1. Determine the real analytic surface

$$\mathcal{C} = \{(x, y, q) > 0 : 0 < q < 1, \partial_q B_q(x, y) = 0\},$$

its number of sheets over the (x, y) -plane, and the cusp or Maxwell loci of its critical values. Establish parameter regions with zero, one, or multiple positive-base inverse branches.

Lambert-series differentiation of Equation (21.1) gives a rapidly convergent exact equation for $\mathcal{C}$; interval continuation can turn the initial numerical atlas into proofs.

25.5 Root-of-unity universality for multinomial inverses

The carry theorem fixes the ramification order but not the full unfolding. Allowing the integer parts to move through q -gamma interpolation produces continuous parameters near the cyclotomic zero.

Conjecture 25.4 (Generic A_{m-1} unfolding). At a primitive d -th root with q -multinomial carry multiplicity m , a generic $(m-1)$ -parameter continuous deformation of the parts gives a versal unfolding of the singularity $y = c(q - \zeta)^m$, equivalent to the A_{m-1} normal form.

This would connect elementary floor carries to the singularity theory of inverse coverings and would predict all local branch-merger diagrams.

25.6 Coefficient arithmetic of the double-scaling inverse

The polynomials $A_m(L)$ in Equation (12.1) are generated by Bernoulli numbers and partitioned compositions. Their denominators and signs show nontrivial structure.

Research problem 25.2. Find a closed formula for $A_m(L)$ in partial Bell polynomials, determine the minimal denominator as a function of m , and study the zero distribution of $A_m(L)/L$. Decide whether a natural Bernoulli rescaling produces integer coefficients or a combinatorial interpretation.

25.7 Fabius–Rvachev inverse consistency

The family Equation (23.1) provides a direct base estimator from each even cumulant. For exact geometric scaling all estimators coincide. A broader inverse problem allows the widths to be $w_j = (1 - q)q^j(1 + \epsilon_j)$ with a small structured perturbation.

Research problem 25.3. Recover finitely many deformation modes $(q, \epsilon_0, \dots, \epsilon_s)$ from a finite moment vector, determine the exact Jacobian rank at the dyadic point, and derive minimax conditioning bounds. Relate loss of rank to the flatness and endpoint asymptotics of the associated Fabius/Rvachev functions.

Good's multivariate formula supplies formal inverse coefficients; the remaining work is global identifiability and analytic error control.

25.8 Formal verification roadmap

The algebraic core is well suited to Lean formalization in the following order.

1. finite formal-series reversion and the triangular recurrence;
2. cyclotomic exponents for Gaussian coefficients, q -multinomials, and q -factorials;
3. the carry theorem and square-free Gaussian corollary;
4. exact $q = -1$ identities, including [Equation \(7.6\)](#) and [Equation \(16.4\)](#);
5. positivity and the centered-palindromic monotonicity theorem;
6. finite-order coefficient identities for the Bernoulli jets;
7. analytic convergence, asymptotic remainder estimates, and sectorial continuation.

The first five items require only polynomial algebra, finite sums, and real monotonicity. They can therefore be separated cleanly from the much heavier complex asymptotic infrastructure.

Part VII

Reference appendices

Chapter 26

Inverse-regime tables

26.1 Finite q-Pochhammer product

Table 26.1: Local inverse types for $(a; q)_n$.

Recovered variable	Base point	Type	Leading coordinate/criterion
a	$a = 0, y = 1$	Taylor	derivative $-[n]_q$; singular when $[n]_q = 0$
a	$a = q^{-j}, y = 0$	Taylor	exact derivative Equation (5.4)
a	$a = \infty$	n -Puisseux	$a \sim (y/((-1)^n q^{\binom{n}{2}}))^{1/n}$
q	$q = 0$	Taylor	normalized target r , Equation (6.1)
q	$q = 1$	Taylor	$\varepsilon = -\text{Log } q$, logarithmic target w
q	$q = -1, a = a_*$	square-root	fold for $n \geq 4$
q	$q = -1, n = 3, a = -1/3$	cubic root	exact Equation (7.6)
q	root ζ , target zero	m -Puisseux	multiplicity count Equation (7.10)
q	$q = \infty$	$\binom{n}{2}$ -Puisseux	reciprocal factorization Equation (8.1)
α	regular complex order	Taylor	derivative Equation (9.3)

26.2 Gaussian and related polynomial q-analogs

Table 26.2: Local inverse types for Gaussian and related polynomials.

Object	Base point	Type	Leading mechanism
Gaussian	$q = 0, y = 1$	Taylor	restricted partitions
Gaussian	$q = 1$, uncentered	Taylor	deterministic drift $-d/2$
Gaussian	$q = 1$, centered	square-root	variance/cumulants
Gaussian	primitive root, target zero	Taylor	square-free cyclotomic factor
Gaussian	$q = -1$, zero case	Taylor	exact derivative Equation (16.4)
Gaussian	$q = \infty$	d -Puisseux	reciprocity
q-multinomial	primitive root	e_d -Puisseux	residue-carry count

Object	Base point	Type	Leading mechanism
q-factorial	primitive root	$\lfloor n/d \rfloor$ -Puisseux	multiples of d
positive palin-	centered $q = 1$	square-root	symmetric moment-generating function
dromic polynomial			

Chapter 27

Coefficient formula compendium

27.1 Scalar ordinary reversion

For $w = c_1 z + c_2 z^2 + c_3 z^3 + c_4 z^4 + \dots$,

$$z = \frac{w}{c_1} - \frac{c_2}{c_1^3} w^2 + \frac{2c_2^2 - c_1 c_3}{c_1^5} w^3 + \frac{-5c_2^3 + 5c_1 c_2 c_3 - c_1^2 c_4}{c_1^7} w^4 + \mathcal{O}(w^5).$$

For implementation, the recurrence in Part I avoids expression swell from closed Bell-polynomial formulas.

27.2 Fold inversion

If

$$y - y_0 = Az^2 + Bz^3 + Cz^4 + \dots, \quad A \neq 0,$$

and $\xi = ((y - y_0)/A)^{1/2}$, then

$$z = \pm \xi - \frac{B}{2A} \xi^2 \pm \left(\frac{5B^2}{8A^2} - \frac{C}{2A} \right) \xi^3 + \mathcal{O}(\xi^4).$$

The signs on odd powers follow the selected square-root sheet.

27.3 Centered cumulant inversion

For $u = \sum_{r \geq 1} \kappa_{2r} \varepsilon^{2r} / (2r)!$, first revert the ordinary series in $z = \varepsilon^2$, then take a square root. This two-stage method is algebraically more stable than direct Puiseux coefficient matching.

27.4 Reciprocal polynomial inversion

If

$$F(q) = Cq^D(1 + c_1 q^{-1} + c_2 q^{-2} + c_3 q^{-3} + \dots), \quad Y = (y/C)^{1/D},$$

then

$$q = Y - \frac{c_1}{D} + \left(\frac{c_1^2(D-1)}{2D^2} - \frac{c_2}{D} \right) Y^{-1} + \left(-\frac{c_1^3(D-1)(2D-1)}{3D^3} + \frac{2c_1 c_2(D-1)}{D^2} - \frac{c_3}{D} \right) Y^{-2} + \mathcal{O}(Y^{-3}).$$

This single template yields [Equations \(8.2\), \(17.2\) and \(19.9\)](#).

Chapter 28

Reproduction notes and source map

28.1 Archive contents

The companion archive contains:

`inverse_q_analog_jet_atlas.tex` the complete LaTeX source;
`inverse_q_analog_jet_atlas.pdf` the compiled report;
`inverse_q_analog_experiments.py` exact and high-precision checks with detailed comments;
`numerical_checks.txt` the deterministic output generated at 80 decimal digits;
`README.txt` build and reproduction commands.

28.2 Principal audited TeX sources

The live source tree at the pinned commit contains, among its TeX material, the following principal documents:

- `Exponents_and_q_Series_Frontiers/ Exponents_and_q_Series_Frontiers.tex`;
- `Fabius_Flat_Parameter_Response_Dynamics/ fabius_frontier_report.tex`;
- `Fabius_Rvachev_Frontier_Report/ fabius_frontier_report.tex` and its generated `numerical_results.tex`;
- `inverse_q_analogs_report/ inverse_q_analogs_report.tex`;
- `q_pochhammer_q_binomial_expansions_report/ q_analog_expansions_report.tex`;
- `q_pochhammer_q_binomial_monograph/ q_pochhammer_q_binomial_monograph.tex`;
- small generated TeX table fragments under the synthesis assets.

The forward identities were cross-checked against the proof-oriented monograph; the new inverse statements were then derived independently and verified where practical by the archive script.

28.3 Build commands

From the archive directory, a standard TeX installation can reproduce the PDF with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \  
inverse_q_analog_jet_atlas.tex
```

The numerical audit is regenerated with

```
python inverse_q_analog_experiments.py --digits 80 \  
    --output numerical_checks.txt
```

The script requires only SymPy and mpmath.

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